

APPLIED MATHEMATICS FOR ENGINEERS - I (25MAT11)

Module -2: Applied Linear Algebra and its properties

Eigen values and Eigen vectors of a square matrix and its standard properties, Rayleigh's power method to find the dominant Eigenvalue and Eigenvector, Power of matrices using Diagonalization. Solutions of homogeneous linear differential equation using matrix method.

Applications | Traffic, Network flow and Electrical circuit problems.

PART-A

1. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$
2. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$
3. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$
4. Find all the characteristic values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$
5. Find the eigen values of the matrix $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$
6. Find the eigen values of the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$
7. Find the eigen values of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$
8. Find the characteristic values of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$
9. Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix}$
10. Find the modal of matrix of $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$
11. Diagonalise the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and hence find A^5
12. Diagonalise the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ and hence find A^4
13. Diagonalise the matrix $A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$

14. Reduce the matrix $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ to diagonal form.

15. Diagonalise the matrix $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$

16. Find eigenvalues and eigenvectors of matrix $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and diagonalize it.

17. Reduce the matrix $A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$ to diagonal form. Also find A^5 .

18. Find the diagonal matrix of $A = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$

19. Is the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ diagnosable?

20. Find the numerically largest eigen value and the corresponding eigenvector of the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \text{ by Rayleigh's power method.}$$

21. Use Rayleigh's power method to find the dominant eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ by taking initial vector as $[1 \ 0 \ 0]^T$.

22. Using Rayleigh's power method find numerically the largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by taking initial vector as $[1 \ 1 \ 1]^T$.

23. Using Rayleigh's power method find numerically the largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ -2 & 1 & 5 \end{bmatrix}$ by taking initial vector as $[1 \ 0.8 \ -0.8]^T$.

24. Find the largest eigen value and the corresponding eigenvector of the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \text{ by power method taking initial vector as } [1 \ 1 \ 1]^T.$$

25. Find the largest eigen value and the corresponding eigenvector of the matrix

$$A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ by Rayleigh's power method taking initial vector as } [1 \ 0 \ 0]^T.$$

26. Use Rayleigh's power method to find the dominant eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$ by taking initial vector as $[1 \ 1 \ 1]^T$.

27. Find the largest eigen value and the corresponding eigenvector of the matrix

$A = \begin{bmatrix} 10 & 2 & 1 \\ 2 & 10 & 1 \\ 2 & 1 & 10 \end{bmatrix}$ by Rayleigh's power method taking initial vector as $[1 \ 0 \ 0]'$.

28. Let A be a 2×2 matrix with eigen values -3 and -1 and corresponding eigen vectors $v_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Let $X(t)$ be the position of a particle at time t . Solve the initial value problem $X' = AX$, $X_0 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$
29. Suppose a particle is moving in a planar force field and its position vector X satisfies $X' = AX$ and $X(0) = X_0$, where $A = \begin{bmatrix} 4 & -5 \\ -2 & 1 \end{bmatrix}$ and $X_0 = \begin{bmatrix} 2.9 \\ 2.6 \end{bmatrix}$. Solve the initial value problem for $t \geq 0$.
30. A particle moving in a planar force field has a position vector X that satisfies $X' = AX$. The 2×2 matrix A has eigen values 4 and 2 with corresponding eigen vectors $v_1 = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Find the position of the particle at time t , assuming that $X_0 = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$.

PART-B

1. For matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$, find eigenvalues and eigenvectors.
2. Find the powers A^{10} and A^{20} of matrix A by diagonalization where $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$
3. Find eigenvectors of $A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ and check if matrix is diagonalizable
4. Reduce the matrix $A = \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix}$ to diagonal form and hence find A^5
5. Reduce the matrix $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$ to diagonal form
6. Reduce the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ to diagonal form.
7. Find the numerically largest eigen value and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ by Rayleigh's power method up to 4 iterations.
8. Find the largest eigen value and the corresponding eigenvector of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by power method taking initial vector as $[0 \ 0 \ 1]^T$
9. Using Rayleigh's power method find the numerically largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ taking initial vector as $[1 \ 1 \ 1]^T$.
10. Using Rayleigh's power method find the numerically largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 10 & 2 & 1 \\ 2 & 10 & 1 \\ 2 & 1 & 10 \end{bmatrix}$ by taking initial vector as $[0 \ 0 \ 1]^T$.

PART- C

1. Find all the Eigen values of a matrix $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$
2. Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$
3. Find the characteristic roots and Eigen vectors of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$
4. Find all the Eigen values of a matrix $\begin{bmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{bmatrix}$
5. Reduce the matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix}$ to diagonal form and hence find A^5
6. Diagonalise the matrix $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$
7. Diagonalise the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$
8. Use Rayleigh's method to find the dominant eigenvalue of matrix: $A = \begin{bmatrix} 4 & 1 & 1 \\ 1 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$
starting with vector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ for 4 iterations.
9. Using Rayleigh's power method find the numerically largest eigenvalue and corresponding eigenvector of the matrix $A = \begin{bmatrix} 1 & 3 & -1 \\ 3 & 2 & 4 \\ -1 & 4 & 10 \end{bmatrix}$ taking initial vector as $[1 \ 1 \ 0]^T$.
10. Use Rayleigh's method to find the dominant eigenvalue of matrix: $A = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$